



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer: The program is designed to manage incident tickets reported by IT automation bots. It allows the user to add new incident tickets, display active tickets, search for a specific incident, remove a resolved ticket, and count the active tickets. The program uses a Python list to store the incident ticket records and applies list operations to manage the active tickets.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident: `tickets.append(new_ticket)`

- Creates a new ticket dictionary from the user's Incident ID, Bot Name, and Short Description, then appends it to the tickets list.

Display Incidents: `for number, ticket in enumerate(tickets, start=1):`

- Loops through the tickets list and displays each active ticket together with its Incident ID, Bot Name, and Short Description.

Search Incident: `for ticket in tickets:`

- Searches through the list and compares the entered Incident ID with each ticket ID. The comparison is case-insensitive.

Remove Resolved Incident: `tickets.remove(ticket)`

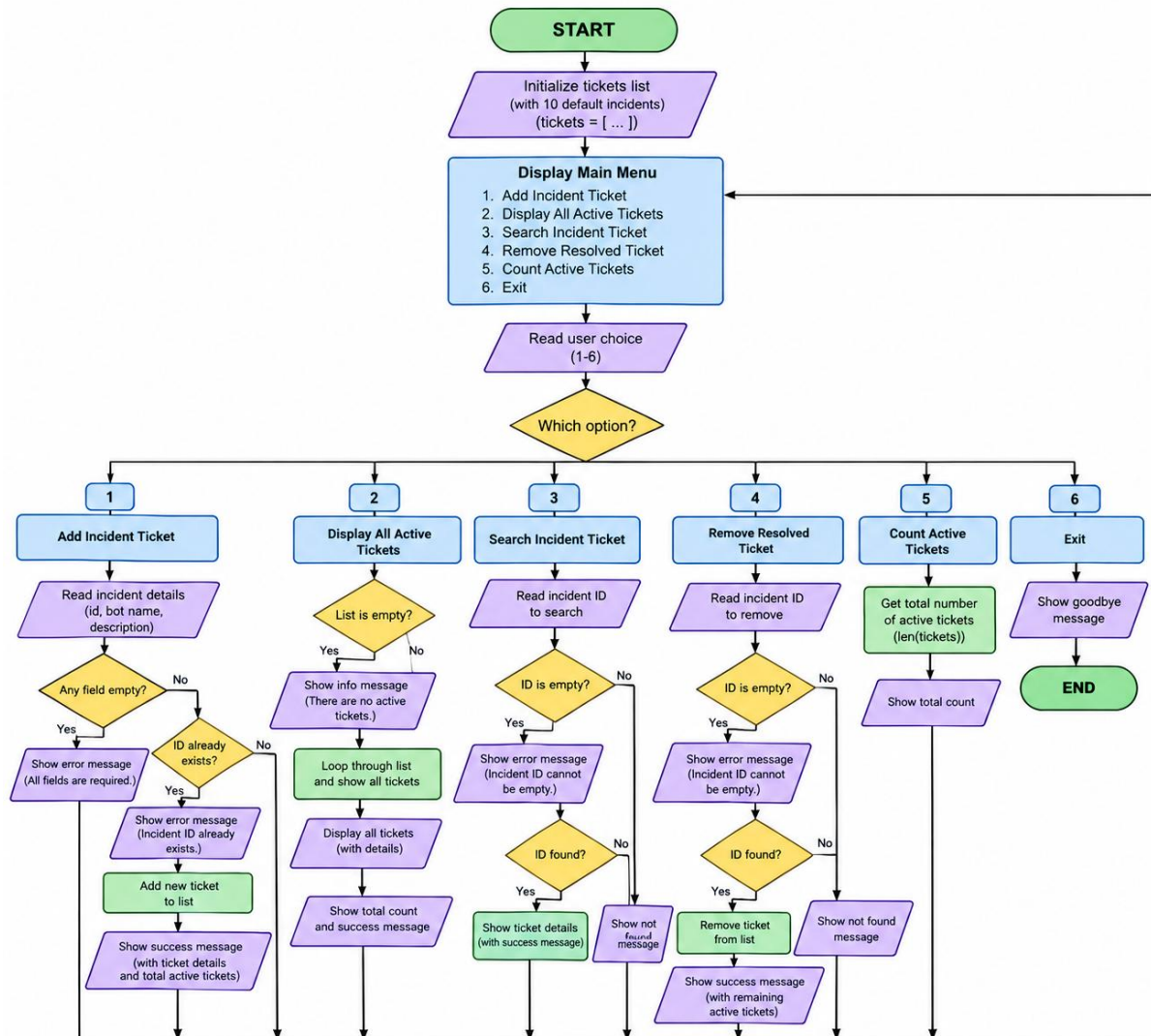
- Removes the matching ticket from the list when an incident has been resolved, so it is no longer included among active tickets.

Count Active Incidents: `total = len(tickets)`

- Uses the length of the list to determine and display the current number of active incident tickets.



III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.



IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

This activity helped demonstrate how a linear data structure can be used in a practical IT incident management scenario. I learned how a Python list can store multiple incident records and how operations such as append, iteration, remove, and len can be used to manage the data. The activity also showed how a menu-driven program can connect these data structure operations into one working application.